

NeuroPulse AI: Electromyography (EMG) Profile

Yellow = cleaned signal | Green = muscle strength | Red = muscle active shadow

Patient: Aditya | **ID:** NP-20260522-6079 | **Date:** 22-05-2026 19:56

Avg Strength 4.9	Active Time 8.4%	Major Spikes 3	Signal Quality Low
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1. Clinical Summary

Session Score	12/100
Symmetry Score	0.8%
Side Deficit	99.2% (Right lower)
Lowest Activity Channel	Channel 3
Fatigue Drop	High (79.5% decline)
Signal Quality	Low

2. 30-Second Graph Snapshot



3. Muscle-wise Quantitative Summary

Muscle / Channel	Avg	Peak	Active %	Quality
Channel 1	18.6	224.4	32.1%	Good
Channel 2	0.8	121.5	1.2%	Low
Channel 3	0.0	0.0	0.0%	Low
Channel 4	0.2	181.8	0.2%	Low

4. Local Data-Based EMG Findings

Signal Reasoning (Quantitative Text Analysis):

- * Session Score: 32/100. Signal Quality: Fair.
- * Side Deficit: 99.2%. Symmetry Score: 0.8%. Marked right-side deficit detected.
- * Lowest Activity: Channel 3. Highest Peak: Channel 1.
- * Fatigue Drop: Low (0.0% decline from early to recent activity).
- * Channel Balance: left-side channel balance 4%; right-side channel balance 0%.
- * Suggested focus: compare the two right-side muscles/channels and train the weaker activation pattern.
- * Selected affected side: Select Affected Side. Non-diagnostic rehab monitoring support only.

Recorded Session Findings:

- * Recording summary: 6420 EMG samples captured over approximately 60 seconds.
- * Session Score: 12/100. Signal Quality: Low.
- * Side Deficit: 99.2%. Symmetry Score: 0.8%. Impression: marked right-side activation deficit. Bilateral EMG symmetry is reduced and requires targeted rehab attention.
- * Lowest Activity: Channel 3 showed the lowest average activation (0.0).
- * Highest Peak: Channel 1 showed the highest peak activation (224.4).
- * Channel Balance: left-side channel balance 4.4%; right-side channel balance 0.0%.
- * Fatigue Drop: High fatigue pattern with 79.5% decline from early to recent activity.
- * Sustained activation: Channel 1 32.1%, Channel 2 1.2%, Channel 3 0.0%, Channel 4 0.2% above threshold.
- * Clinical note: Affected side was not selected; interpretation is limited to channel activity and symmetry. This is a non-diagnostic rehab monitoring prototype that provides quantitative EMG feedback for physiotherapy support.

5. Clinical Observations & Notes

Summary Notes: N/A

Comparative Analysis:

- Left Side Avg: 9.7
- Right Side Avg: 0.1
- Symmetry Ratio (L/R): 0.8%
- Most Dynamic Channel: Channel 1